

II-Probabilistic Approaches Causality

Foundations of Neuro-Symbolic AI

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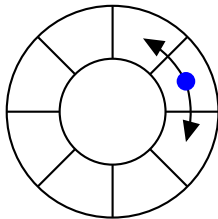
Outline

- ▶ Ambiguity in Bayesian Networks
- ▶ Modelling interventions
- ▶ Reasoning about hypothetical worlds: Counterfactuals

Example: Random walk

Imagine a random walk through circular cells:

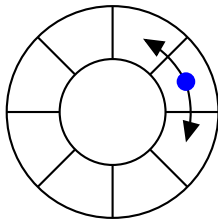
- ▶ X_k describes at timestep $k \in [d]$ the position
- ▶ With probability 0.5 we stay at a cell, and to each neighboring cell with 0.25



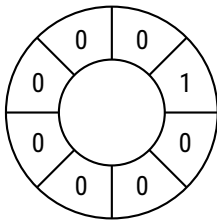
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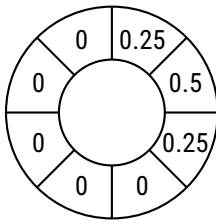
- ▶ X_k describes at timestep $k \in [d]$ the position
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Example: When observing the state 1 for X_k we transit as



$$\mathbb{P}[X_k] = e_1[X_k]$$



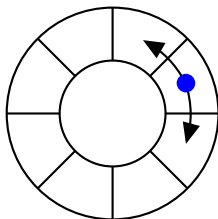
$$\mathbb{P}[X_{k+1}] =$$

$$(0.25 \quad 0.5 \quad 0.25 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0) [X_{k+1}]$$

Example: Random walk

Imagine a random walk through circular cells:

- ▶ X_k describes at timestep $k \in [d]$ the position
- ▶ With probability 0.5 we stay at a cell, and to each neighboring cell with 0.25



This is a Markov chain with the transition matrix:

$$\mathbb{P}[X_{k+1}|X_k] = \begin{pmatrix} 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 \\ 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 \\ 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 \end{pmatrix} [X_{k+1}, X_k]$$

Reasoning about the future and the past

Intuitive: Conditioning on the past we reason on the future:

$$\begin{aligned}\langle \epsilon_{X_k} [X_k], \mathbb{P}[X_{k+1}|X_k] \rangle_{[X_{k+1}]} &= 0.5 \cdot \epsilon_{X_k} [X_{k+1}] \\ &\quad + 0.25 \cdot \epsilon_{X_k-1} [X_{k+1}] + 0.25 \cdot \epsilon_{X_k+1} [X_{k+1}]\end{aligned}$$

Counter-intuitive: Conditioning on the future we can also reason on the past:

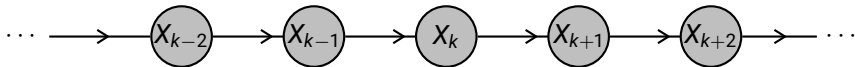
$$\begin{aligned}\langle \mathbb{P}[X_{k+1}|X_k], \epsilon_{X_k} [X_{k+1}] \rangle_{[X_k]} &= 0.5 \cdot \epsilon_{X_k} [X_k] \\ &\quad + 0.25 \cdot \epsilon_{X_k-1} [X_k] + 0.25 \cdot \epsilon_{X_k+1} [X_k]\end{aligned}$$

How to distinguish between these directions?

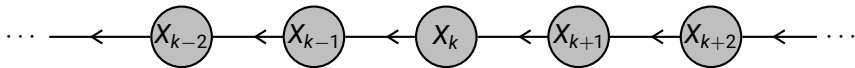
How to distinguish the past from the present?

We have two decomposition schemes as Bayesian networks, both decorated with the same transition matrices:

Parametrization from the past to the future:



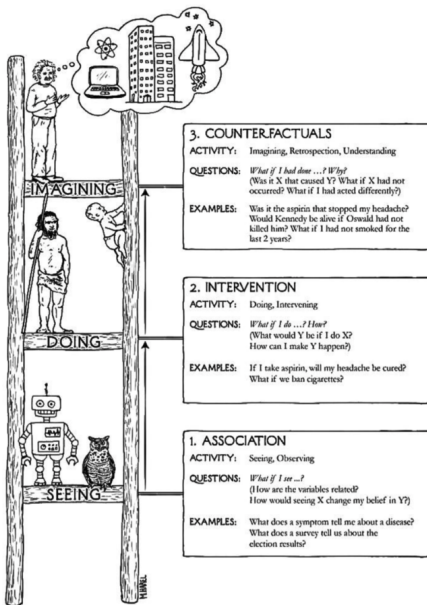
Parametrization from the future to the past:



Ambiguity in Bayesian Networks

Causation is not (always) reflected in the directions.

The Ladder of Causality



→ Model randomness by
response variables L_k
and add
counterfactual variables X_k

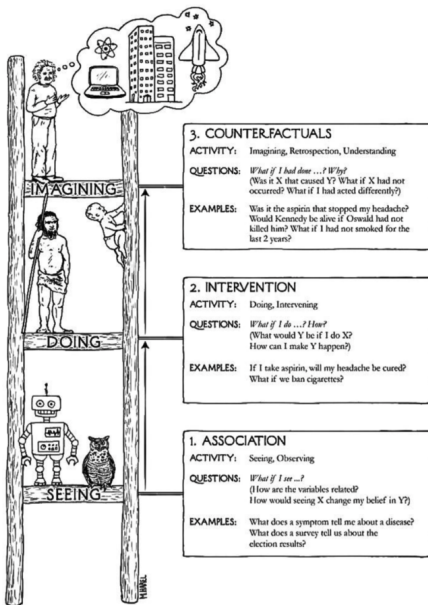
→ Model interventions by
do variables D_k

→ Bayesian Networks
model the distribution of
observed variables X_k

Outline

- ▶ Ambiguity in Bayesian Networks
- ▶ **Modelling interventions**
- ▶ Reasoning about hypothetical worlds: Counterfactuals

The Ladder of Causality



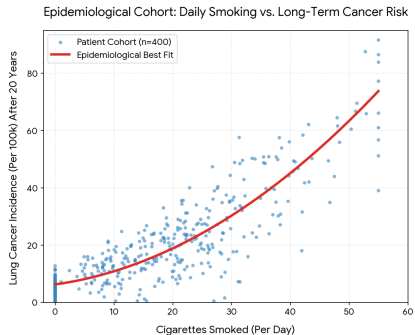
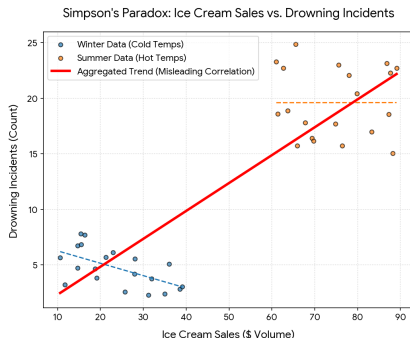
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Bayesian Networks
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Correlation does not imply causation

Consider two examples¹ of correlations:



Question: How to reduce casualties based on public laws and taxes?

¹generated data

Interventions

Interventional reasoning

AI is not only about modelling the observed, but reasoning about consequences: When we intervene on a variable, what would be the effect on the further variables?

This is necessary in many realistic situations:

- ▶ Will product sales increase if we lower the price?
- ▶ Will I pass the exam tomorrow if I study today?

Idea:

- ▶ Understand the directed hyperedges to model causal consequences.
- ▶ But: How to deal with the ambiguity?

Associations in Bayesian Networks can not model interventions!

Do variables: Modelling interventions

We add to each variable X_k a **do variable** D_k of dimension $[m_k + 1]$ modelling the intervention on it:

- ▶ Intervention on the variable X_k when $d_k \in [m_k]$
- ▶ No intervention on the variable when $d_k = m_k$

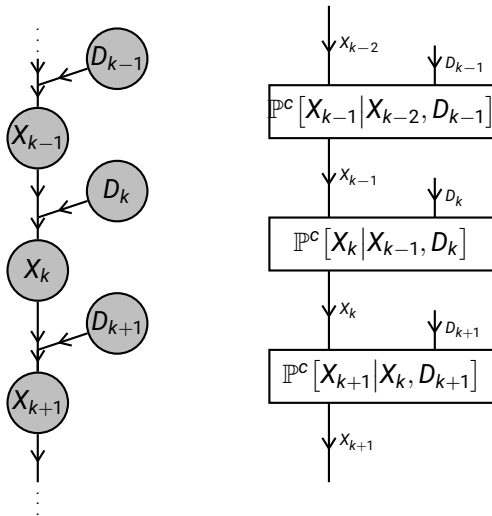
This is realized by a tensor

$$\begin{aligned}\mathbb{P}^c[X_k | X_{\text{Pa}(k)}, D_k] &= \sum_{x_k \in [m_k]} \epsilon_{x_k}[X_k] \otimes \mathbb{I}[X_{\text{Pa}(k)}] \otimes \epsilon_{x_k}[D_k] \quad (\text{intervention}) \\ &+ \mathbb{P}[X_k | X_{\text{Pa}(k)}] \otimes \epsilon_{m_k}[D_k] \quad (\text{no intervention})\end{aligned}$$

This resolves the ambiguity in Bayesian network representation!

Causally augmented Markov chain

For a Markov chain we get the causally augmented model:



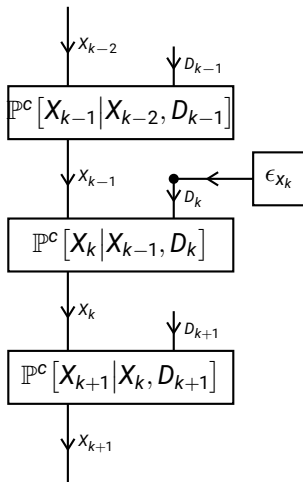
Example: Interventions on a random walk

The causally augmented conditional distributions are for the random walk with $m = 3$:

$$\mathbb{P}^c[X_k | X_{k-1}, D_k] = \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}}_{\text{Intervention}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \overbrace{\begin{pmatrix} 0.50 & 0.25 & 0.25 \\ 0.25 & 0.50 & 0.25 \\ 0.25 & 0.25 & 0.50 \end{pmatrix}}^{\text{No intervention}} [X_k, X_{k-1}, D_k]$$

Intervening on a position

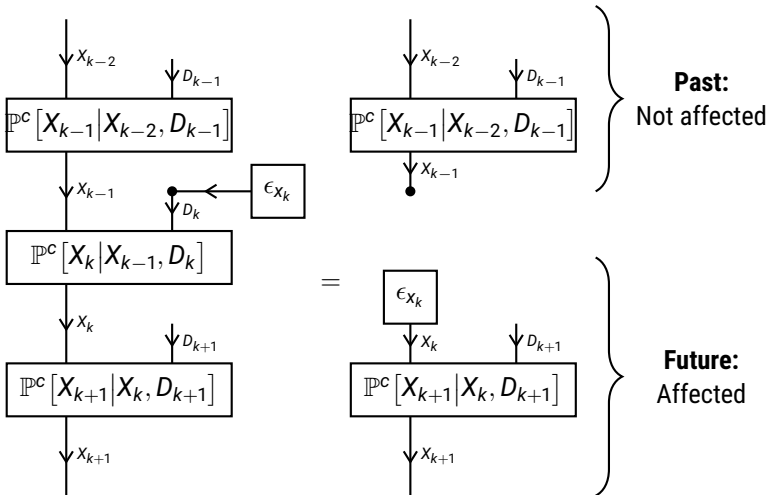
Query: What is the consequence of intervening at time k in state x_k ?



Interpretation: Restart of the random walk at time k in state x_k .

Intervening on a position

Query: What is the consequence of intervening at time k in state x_k ?



Interpretation: Restart of the random walk at time k in state x_k .

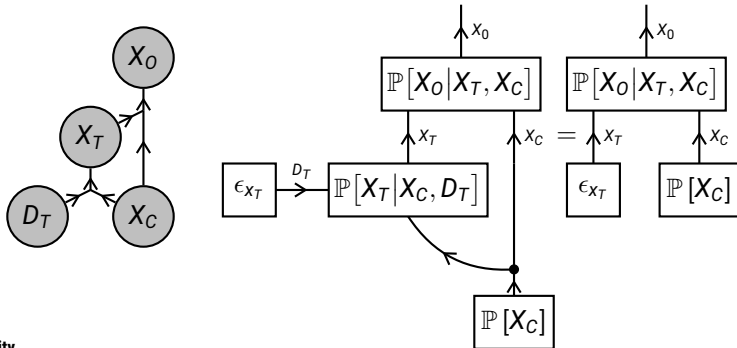
Randomized controlled trial

Aim

Model the causal effect of a treatment X_T (e.g. a drug) on an observation X_O (e.g. patients health).

- How to assign the treatment to the patients?
- How to determine the causal effect?

Naive approach: Voluntarily treatment and observation of the effects.



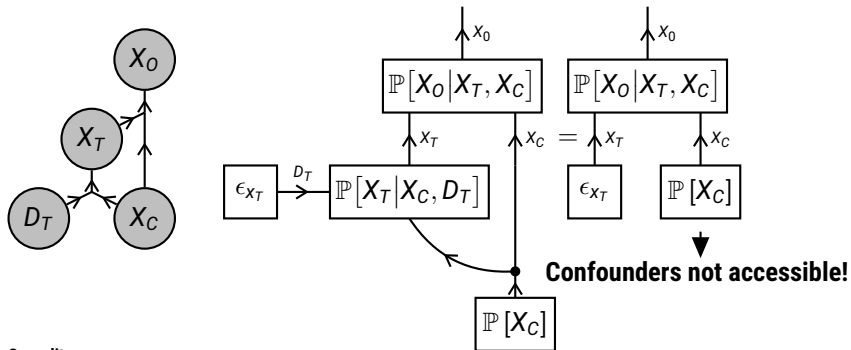
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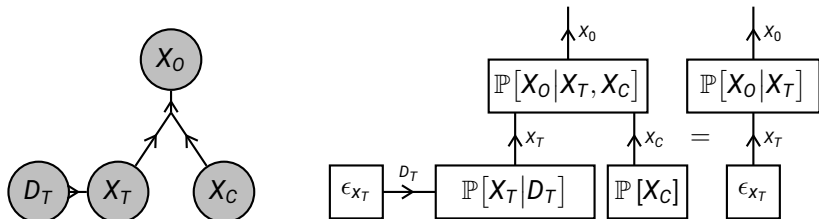
Randomized controlled trial

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- How to assign the treatment to the patients?
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Randomized controlled trial: Assign treatment **independently** to test group

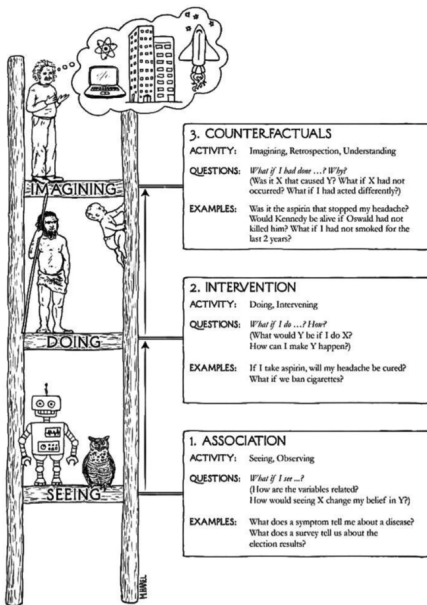


Influence is identifiable by conditional queries!

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Counterfactuals

Counterfactual reasoning

- ▶ A particular state has been observed.
- ▶ We reason about a hypothetical world.

This appears in important situation:

- ▶ Would the accident have happened had the safety protocol been respected?
- ▶ Would the person have developed cancer had he stopped smoking earlier?

Interventions are not enough for these questions!

Response variables

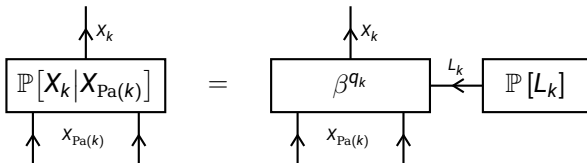
Interpretation of randomness

Random behavior in observed variables is due to exogenous variables, which states are not observed.

To decompose each conditional distribution we introduce:

- ▶ Response variables L_k summarize the exogenous variables
- ▶ Function q_k describing a deterministic dependence of x_k on the parent states $x_{\text{Pa}(k)}$ and the state ℓ_k of the response variable

The conditional probabilities are then contracted basis encodings, that is



Random walk example

We model a response variable

- ▶ $\ell_k = 0$: going to the left cell
- ▶ $\ell_k = 0$: staying at the current cell
- ▶ $\ell_k = 2$: going to the right cell

$$\mathbb{P}[X_k | X_{k-1}, L_k] = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} [X_k, X_{k-1}, L_k]$$

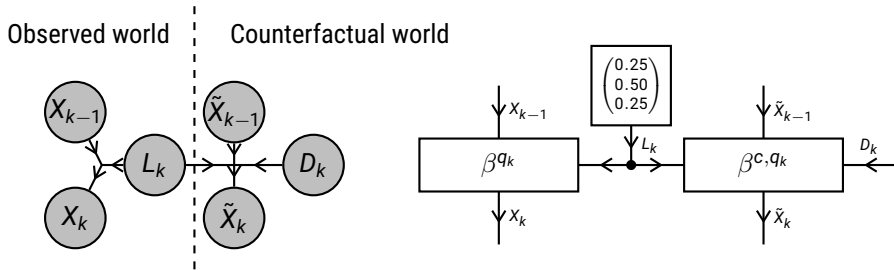
Effect in random walk: Shift of the random walk.

Counterfactual variables

Total setup to reason about interventions on a hypothetical world:

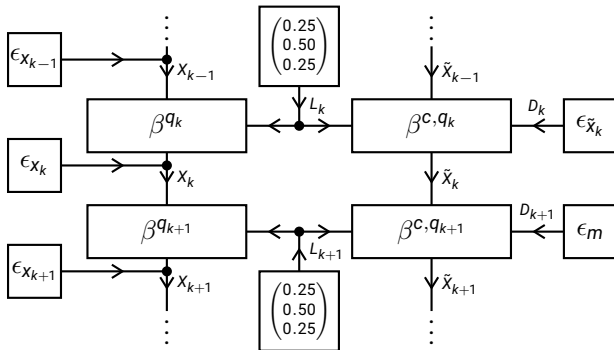
- ▶ Observed variables X_k : describing the observed world
- ▶ Response variables L_k : summarizing exogenous variables
- ▶ Counterfactual variables $\tilde{X}_k$: describing an hypothetical world
- ▶ Do variables D_k : interventions in the hypothetical world

In the random walk example:



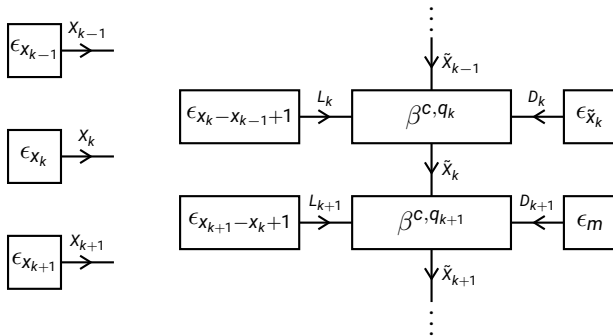
Counterfactual query

What would be the outcome of the walk, would we have done the intervention on X_k ?



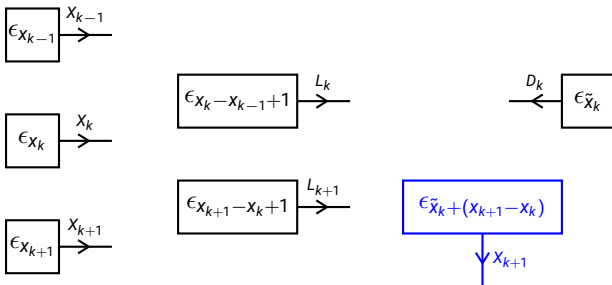
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Answer: The walk would be shifted by the intervention. Different to the intervention query, we have a deterministic outcome!